Gregory Bian

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Education

University of British Columbia Expected May 2028
Bachelor of Applied Science, Computer Engineering *Vancouver, BC*

Work Experience

Geotab Burnaby, BC
Embedded Software Developer Intern – Vehicle Platform Development *May 2026 – Present*

- Added firmware support in C to 15+ OEM vehicles for Geotab's telematics platform and customer-base.
- Integrated 30+ vehicle signals and VIN decoding using OBD-II, UDS over ISO-TP, J1939 CAN protocols, giving fleet customers EV battery, range, speed, and odometer data on newly supported vehicles.
- Cut stale feature flag removal effort by 90% by automating code edits, PRs, and CI/SIL monitoring end-to-end with Claude Skills on VMs; removed 20+ flags and adopted the process team-wide.

Engineering Design Teams and Extracurriculars

UBC Solar Racing Vancouver, BC
Embedded Systems Team Lead *Jun 2026 – Present*

- Lead a 9-person subteam within an 80+ member team, delivering firmware and telemetry for a 4-ECU solar car.
- Plan milestones for a two-year design cycle, directing design and code reviews and mentoring members.
- Architected SIL and Regression testing pipelines using Renode hardware emulation and Robot testing Frameworks to simulate RTOS and hardware edge cases.

Senior Designer *Apr 2026 – Jun 2026*

- Led end-to-end integration tests to verify telemetry pipelines over radio/cellular, safety features for motor control, and integration with the BMS team, ensuring reliable control over all safety-critical systems.
- Developed automated Bash and Python testing scripts to improve the reliability and coverage of integration tests.

Embedded Firmware Developer *Sep 2025 – Apr 2026*

- Architected the multi-page LCD dashboard firmware, displaying 5x more vehicle data than previous iterations for faster fault detection and real-time driver awareness.
- Cut live vehicle telemetry packet loss by 30% by writing UART drivers for an XBee radio, using RTOS semaphores to synchronize shared telemetry buffers across concurrent tasks.
- Diagnosed 5+ hardware issues on custom PCBs by probing signals with oscilloscopes, multimeters, and PCANs.

nwPlus Vancouver, BC
Engagement Coordinator *May 2026 – Present*

- Fostering a collaborative team culture across a 50+ member organizing committee to deliver the largest Canadian Northwestern University hackathon, giving 600+ participants a platform to showcase their technical skills.

Technical Skills

Programming & Frameworks: C, C++, Python, Java, JavaScript (React), Bash, ARM Assembly
Embedded Systems: FreeRTOS, STM32, CAN (OBD-II, UDS, ISO-TP, J1939), UART, SPI, I2C
Tools: Claude Code, Linux (Ubuntu), Git, J-Link, STM32CubeMX, CMake, Grafana, InfluxDB, SavvyCAN